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Transient dynamics of three-dimensional beam trusses under impulse loads

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Abstract: Spatial structures are often subjected to impulse loads which induce high-frequency (HF) wave propagations. Despite some recent researches, the characterization of the transient response to such loads remains an open problem. The objective of this research is to develop a reliable model of the HF energy evolution within three-dimensional Timoshenko beam trusses in order to predict, for example, their potential steady-state behavior at late times or the energy paths. The theory of micro-local analysis of wave systems shows that the energy density associated with their HF solutions satisfies a Liouville-type transport equation. At the interfaces between substructures, the energy flow is partly reflected and partly transmitted. The corresponding power flow reflection/transmission coefficients are also derived in this study. Numerical simulations are performed by a spectral Discontinuous "Galerkin" (DG) method for spatial discretization and a strong stability-preserving Runge-Kutta (RK) method for time integration. Numerical results using the RK-DG method are presented for the example of a three-dimensional beam truss that exhibits a diffusive behavior at late times.
Keywords: high frequency, transport equation, beam model.

1. Introduction

In this research, we are interested in the high-frequency (HF) wave propagation within three-dimensional beam trusses. In this range, classical finite element methods are not applicable on account of the small wavelengths and high modal densities. Therefore several heuristic strategies have been developed by engineers. Global approaches like the Statistical Energy Analysis (SEA) [Lyon and DeJong, 1995] of structural-acoustic systems give information on the mean energy level in each substructure. The main issue in SEA is the estimation of some core parameters like the coupling loss factors or the power inputs. Local approaches like the Vibrational Conductivity Analogy (VCA) [Nefske and Sung, 1989] have been also investigated. The point of issue concerning the existing local approaches is their extension to built-up structures. The approach adopted here is to consider a transport formulation for the energy density carried by HF waves. Indeed, thanks to the microlocal analysis of classical wave systems, it can be proved that the energy density associated with HF solutions of classical wave systems satisfies a Liouville-type transport equation. This theory can be adapted to quantum wave, classical acoustic or electromagnetic waves as shown by Papanicolaou and Ryzhik (1999).

Transport equations can be solved numerically by various methods including Monte-Carlo methods [Lapeyre et al. 2003] or nodal/spectral Discontinuous "Galerkin" (DG) finite element methods

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[Hesthaven and Warburton, 2008]. Both techniques have the advantage to be weakly dissipative and weakly dispersive (depending on the order of interpolation in DG methods). Thus they are well suited for late-time simulations in order to exhibit the diffusive behavior of the transport equations. These methods have been applied successfully to assemblies of two-dimensional beams or plates [Savin, 2007], as well as to a randomly heterogeneous shell [Savin, 2004].

The main objective of this research is to be able to predict, for example, the diffusive behavior at late times or the energy paths within three-dimensional beam trusses. In the next section of this paper, the mechanical model of a three-dimensional Timoshenko beam is outlined and his transient HF behavior is derived within the transport theory. In order to extend the theory to HF wave propagation in multibay trusses, the various reflection/transmission phenomena at junctions of beams are also briefly analyzed in this section. The resulting equations are numerically integrated by a discontinuous finite element method. This point is addressed in the third part of the paper. The last section offers a few conclusions.

2. HF behavior of Timoshenko beam trusses

2.1 Mechanical modelisation of an isolated beam

The vibrational behavior of the beam is modeled by Timoshenko's beam kinematics. It includes the effects of rotary inertia and shear distortion. Another important point in this theory is the introduction of a shear coefficient to account for the variation of the shear stress along the cross-section. The latter is assumed to be geometrically symmetric such that its geometrical center is on the neutral fiber of the beam. In this paper, beam material is assumed to be isotropic and its heterogeneities vary slowly in regard to the considered wavelengths. A material point $\boldsymbol{s}$ of the beam is parametrized such that $\boldsymbol{s} = s\boldsymbol{t} + s_2\boldsymbol{n} + s_3\boldsymbol{b}$ where $\{s_2, s_3\}$ are the coordinates of the point on the cross-section Σ described by the orthonormal basis vector $\boldsymbol{n}$ and $\boldsymbol{b}$, and s is its coordinate on the neutral fiber C oriented by the tangent vector $\boldsymbol{t}$. The Timoshenko kinematical assumption is:

$$\boldsymbol{U}(s,t) = \boldsymbol{U}_c(s,t) + \boldsymbol{\theta}(s,t) \times \boldsymbol{s}_\perp, \tag{1}$$

where $\boldsymbol{s}_\perp = (0, s_2, s_3)^T$, the displacement of the neutral fiber is $\boldsymbol{U}_c(s,t)$, and $\boldsymbol{\theta}(s,t)$ is the small rotation vector of the cross-section.

Under the assumption of small perturbations, and introducing the vector $\boldsymbol{X}(s,t) = (\partial_t \boldsymbol{U}_c, \partial_t \boldsymbol{\theta}, \boldsymbol{F}, \boldsymbol{M})^T$ in $\mathbb{R}^{12}$ with $\boldsymbol{F}(s,t)$ and $\boldsymbol{M}(s,t)$ respectively the resultant forces and resultant moments acting on the cross-section, the dynamical equilibrium and the constitutive equations of a rigid cross-section within a three-dimensional beam can be written in the form of a hyperbolic problem:

$$\begin{aligned} \boldsymbol{A}(s)\partial_t \boldsymbol{X} &= \boldsymbol{P}(s,\partial_s)\boldsymbol{X}, \\ \boldsymbol{X}(0,s) &= \boldsymbol{X}_0(s), \end{aligned} \tag{2}$$

where $\boldsymbol{A}(s,t) = diag(\rho S \boldsymbol{Id}_3, \boldsymbol{J}, (S\boldsymbol{C}_1)^{-1}, (S\boldsymbol{C}_2)^{-1})$ with $\rho(s)$ the material volume density, S the cross-section area, $\boldsymbol{J}(s,t) = diag(I(s), J(s), K(s))$, with J, K, I the inertias of the beam, $\boldsymbol{C}_1(s) = diag(E, \kappa\mu, \kappa\mu)$ the translational relaxation tensor, with $\kappa\mu(s)$ the reduced shear modulus, and $E(s)$ the Young modulus, $\boldsymbol{C}_2(s) = diag(\kappa\mu I, EJ, EK)$ is the relaxation tensor for rotational motions, and $\boldsymbol{P}(s,\partial_s)$ is the operator given by:

$$\boldsymbol{P}(s,\partial_s) = \begin{pmatrix} 0 & \boldsymbol{A} \\ \boldsymbol{A}^T & 0 \end{pmatrix}, \ \boldsymbol{A} = \begin{pmatrix} \boldsymbol{Id}_3\partial_s & 0 \\ \boldsymbol{\Omega} & \boldsymbol{Id}_3\partial_s \end{pmatrix}, \ \boldsymbol{\Omega} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \tag{3}$$

where $\boldsymbol{Id}_3$ is the 3×3 identity matrix. Details can be found in [Le Guennec and Savin, 2011]. Here $\boldsymbol{X}_0(s)$ is the vector of initial conditions. This system is suitable for studying the energy propagation because the energy density is $\mathcal{E}(s,t)=1/2\boldsymbol{X}^t\boldsymbol{A}\boldsymbol{X}$.

2.3 HF wave propagation in a three-dimensional beam

HF propagation corresponds to a highly oscillatory wave in space, which is very difficult to describe by the classical displacement wave equation. Contrarily, the energy wave is much smoother and will be the quantity of interest for our model. In this context, it is possible to show that the HF limit of the energy density for a given cross-section at an abscissa s is expressed as:

$$\mathcal{E}(s,t)=\sum_{\alpha\in E}\int_{\mathbb{R}} w_\alpha(s,k,t)dk, \tag{4}$$

where $E=\{P_i,T_i,1\le i\le 3\}$ is the energy modes obtained from the eigen decomposition of the dispersion matrix of (3). P_i denotes the compressional modes gathering the longitudinal and the two bending modes, while T_i denotes the transverse modes gathering the torsional and the two shear modes. Their velocities are respectively $c_P=(E/\rho)^{1/2}$ and $c_T=(\kappa\mu/\rho)^{1/2}$.

The power flow density vector $\boldsymbol{\Pi}=\Pi\boldsymbol{t}$ is also useful to investigate on the propagation of the energy density. In the HF range, it is given by:

$$\Pi(s,t)=\sum_{\alpha\in E}\int_{\mathbb{R}} \pi_\alpha(s,k,t)dk, \tag{5}$$

where $\pi_\alpha=c_\alpha(s)w_\alpha(s,k,t)\tilde{k}$, with k the wave number and $\tilde{k}$ the direction of the propagation i.e. the sign of the wave number. Finally it is possible to show that the evolution of the so-called specific intensities w_α is described by the following transport equations:

$$\partial_t w_\alpha+c_\alpha\tilde{k}\partial_s w_\alpha-|\boldsymbol{k}|\partial_s c_\alpha\partial_k w_\alpha=0,\ \alpha\in E. \tag{6}$$

2.2 Reflection/transmission phenomena

The HF energy density propagation in a beam is thus characterized by the transport equations (6). In order to use this model for complex three-dimensional beam trusses, the power flow reflections/transmissions at the junctions have to be considered. The specific intensities w_α do not interfere in the beam, and thus junctions are the only opportunities to couple energetic modes and spread the energy over the truss. These phenomena are analyzed through the reflection/transmission power flow coefficients of a junction $\tau_{\alpha\beta}^{1q}$, for an arbitrary number N of beams, where α is the energetic mode impinging the junction in the beam #1 and β is the transmitted or reflected (if $q=1$) energetic mode in the beam #q. They are computed from the continuity of the displacement, rotation, forces and moments at the junctions. It is observed by computation that, in the HF limit, no rotational (respectively translational) motions are created when a translational (respectively rotational) wave impinges the junction.

3. Numerical simulation for a truss

At the junction, power flows are partly reflected and partly transmitted, resulting in discontinuities of the energy density field. Thus classical finite element method can not be used to solve numerically the Liouville equations (6). A DG finite element method [Hesthaven and Warburton, 2008] is used to solve Equation (6) with reflection/transmission boundary conditions. The theoretical background of the DG method is recalled in section 3.1 below. An illustrative example is presented in a subsequent part.

3.1 Discontinuous "Galerkin" finite elements procedure

Let us consider the subdivision $T_h = \bigcup_{r=1}^{K} D_r$ of S into K non-overlapping subdomains, or finite elements D_r. These elements are written $D_r = [s_{r-1/2}, s_{r+1/2}]$ and the largest element size of this mesh is h. It is assumed that the velocities c_α are constant over the elements; they are denoted by c_α^r. Here the w_α's are considered as functions of $\tilde{k}$ rather than k. A globally defined test space V_h is introduced as $V_h = \oplus_{r=1}^{K} V_h^r$ where the locally defined test spaces $V_h^r \subseteq H_1(D_r)$ do not enforce any particular boundary conditions on ∂D_r. It is thus possible to introduce a numerical flux $\boldsymbol{\pi}_\alpha^*$ which depends on the traces of w_α from both sides of an interface between elements. This leads to the variational formulation:

$$\int_{D_r} (\dot{w}_\alpha v(s) - c_\alpha^r \tilde{k}\, w_\alpha \partial_s v) ds = -[\boldsymbol{\pi}_\alpha^* . \boldsymbol{t}^r v]_{s_{r-1/2}}^{s_{r+1/2}}, \forall v \in V_h. \tag{7}$$

The numerical flux $\boldsymbol{\pi}_\alpha^*$ at the interface $s = s_{r+1/2}$ is constructed from the computed values of $w_\alpha(s_{r+1/2}, k, t)$ in V_r^h (the trace of w_α at $s_{r+1/2}$ in D_r) and $w_\alpha(s_{r+1-1/2}, k, t)$ in V_{r+1}^h (the trace of w_α at $s_{r+1/2}$ in D_{r+1}) so as to be consistent and conservative.

Now a junction of N beams is considered. I_r is the set of indices of the beam elements which are connected to the beam element $\# r$. The numerical fluxes ensuring the stability and consistency of the DG method for transport equations have the form:

$$\boldsymbol{\pi}_\alpha^*(s_{r+1/2}, k, t).\boldsymbol{t}^r = \sum_{\beta \in E} \sum_{r' \in I_r} \tau_{\alpha\beta}^{rr'} \boldsymbol{\pi}_\beta^{r'}(s_{r'+1/2}, k_\beta^{r'}, t).\boldsymbol{t}^{r'}, \tag{8}$$

for $\tilde{\boldsymbol{k}}.\boldsymbol{t}^r < 0$, and:

$$\boldsymbol{\pi}_\alpha^*(s_{r+1/2}, k, t).\boldsymbol{t}^r = \boldsymbol{\pi}_\alpha^r(s, k, t).\boldsymbol{t}^r, \tag{9}$$

for $\tilde{\boldsymbol{k}}.\boldsymbol{t}^r > 0$. This choice corresponds to an upwind numerical flux. The approximation set V_h is constructed as follows:

$$V_h = \{v|_{D_r} \in \mathbb{P}_l(D_r), 1 \le r \le K\}, \tag{10}$$

where $\mathbb{P}_l(D_r) \subset H_1(D_r)$ is typically a polynomial set. In our numerical simulations, $\mathbb{P}_l(D_r)$ is spanned by a set of normalized Jacobi polynomials up to the order l. The numerical dispersion and dissipation errors introduced by this spatial discretization are $O(K^{2l+3})$ and $O(K^{2l+2})$, respectively [Ainsworth, 2004], where $K = h|\boldsymbol{k}|$. These so-called "superconvergence" properties are very much desirable for long time simulations in order to exhibit the diffusion limit of the transport equations (6) with the reflection/transmission boundary conditions. Finally, time integration of the semi-discretized system (8) is performed by a strong stability-preserving high-order Runge-Kutta (RK) scheme [Gottlieb and Shu, 2001].

3.2 Numerical example

A beam truss constituted by $N = 8$ three-dimensional beams is considered: its shape is structured like a pyramid with a rectangular basis having side lengths of $L = 5$ unit length ($L_p = 5$ for $p = 1,3$) or $L = 4$ unit length ($L_p = 4$ for $p = 6,2$). The beam numbered $p = 4$ is $L_4 = 5$ unit length long and the beam numbered $p = 5$ is $L_5 = 3.6$ unit length long (see Figure 1). The other geometrical properties

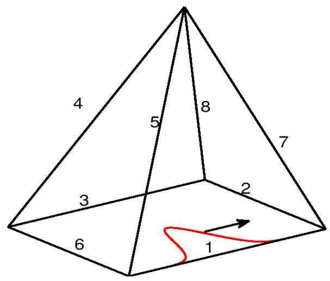

Figure 1- A view of the beam truss with the initial condition on the beam #1 (red line).

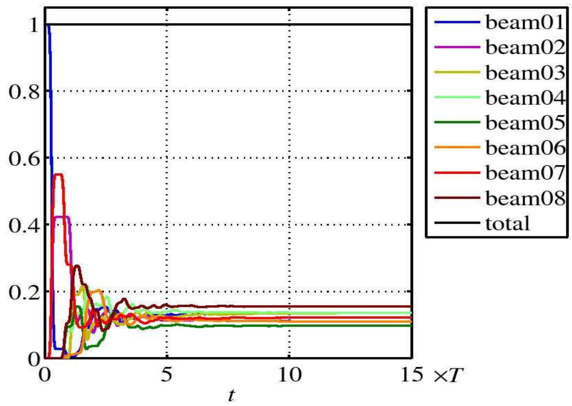

Figure 2-Evolution of the integrated vibrational energy.

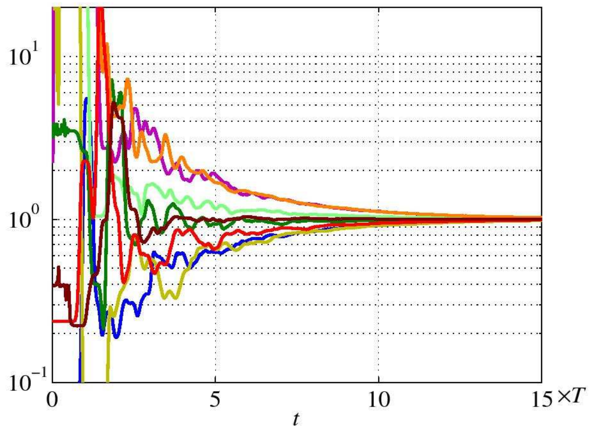

Figure 3-Evolution of ratio $c_T \mathcal{E}_T^p(t)/(2c_P \mathcal{E}_P^p(t))$.

can be inferred from this information. All beams have the same cross-sections of area S and are made from the same homogeneous, isotropic elastic material. The Poisson ratio is $\upsilon = 0.3$ and the shear reduction factor is $\kappa = 5(1+\upsilon)/(6+5\upsilon)$. The energetic initial condition has a Gaussian shape and loads the mode P_1 (a pure compressional wave). It is applied to, say, beam labeled #1 with a fixed initial direction $k = 1$. It is displayed with a red line on the Figure 1. The nodal density is fixed at $20/L$ uniform spatial elements per unit length for all beams. Legendre polynomials up to the fifth order ($l = 5$) are used as local basis functions, and the integration in time is performed by a fourth-order Runge-Kutta scheme. The time scale $T = L_1/c_T$ is introduced: it is the time needed by a transverse wave to travel across the beam #1 of the pyramid. This parameter has been set to $T = 15$ in our simulations. The Courant number $CFL := c_T \Delta t / h$ has been fixed to $CFL = 0.01$. Figure 2 shows the evolution of the total energy for each beam. It should be observed that the total energy for the entire truss is virtually conserved; this is a required property of the numerical scheme retained (the theoretical numerical dissipation error being about 4.10^{-12}). Moreover, the total energy in each beam tends to a limit as $t \to \infty$ which apparently depends on its mechanical parameters and the length of the beam (because we are interested in the integrated energy density over the beam length). This phenomenon exhibits the diffusive regime holding at late times. Finally, Figure 3 shows the ratio between the overall transverse and longitudinal energy per beam $\mathcal{E}_T^p(t)/\mathcal{E}_P^p(t)$, where:

$$\mathcal{E}_\alpha^p(t) = \sum_{j=1}^{3} \sum_{\tilde{k}=\pm 1} \int_{L_p} w_{\alpha_j}(s,\tilde{k},t)ds, \ 1 \le p \le 8, \ \alpha = P,T \tag{11}$$

This ratio converges to a constant value of the form $n_T^p c_P / (n_P^p c_T)$, where n_P^p and n_T^p are the numbers of compressional and transverse modes generated in each beam by the reflection/transmission processes at the junctions. n_P^p and n_T^p depend on the initial condition and the shape of the truss. For example, the initial motion in Figure 2 and Figure 3 is a compressional mode $\alpha = P_1$ in beam #1; it is transmitted as transverse shear modes and longitudinal mode in the beams attached to it (beams #2 and #7); however no conversion to bending or torsional modes occurs at any time. Thus $n_P = 1$ and $n_T = 2$ in this case.

4. Conclusions

A transient transport model for the propagation of the HF vibrational energy density in three-dimensional beam trusses has been derived. It accounts for the power flow reflection and transmission phenomena at the junctions in this range. The proposed analytical model is integrable by a Runge-Kutta Discontinuous "Galerkin" (RKDG) finite element scheme. Owing to its low numerical dispersion and dissipation errors, the RKDG method is suitable for long time simulations. Numerical results obtained for an example of a beam truss illustrate the transient regime and the onset of a diffusive regime at late times. The latter is prompted by the multiple reflections and transmissions of the energy modes at the beam junctions, with possible mode conversions. This scattering process is different from the multiple scattering of waves on random heterogeneities or inclusions considered elsewhere (see [Savin, 2004] or [Papanicolaou, 1999]), but it gives rise to a similar qualitative and quantitative behavior at late times. Diffusion is characterized by an equilibration of the total energy in each subsystem (a beam of the truss) and for each propagation mode, a situation comparable to the assumption of modal equipartition invoked in the Statistical Energy Analysis (SEA) of structural-acoustic systems.

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